EnergyLens — Single-Source Technical & Domain Reference

Everything in the report and the app, explained from first principles (EN) with short Turkish study tips

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Contents

1 How to use this reference

This is your **single source of truth** for everything EnergyLens shows — every formula, every regulation, every term that appears in the report or the app. It is written in **English**, because you must be able to use these terms, standards and certifications in English in front of Dr. Chelabi and future clients. After each idea there is a short **Turkish tip** marked `tr:` to lock the meaning in.

> **tr:** Bu doküman tek kaynağın. İngilizce öğreniyorsun çünkü terimleri/sertifikaları İngilizce bilmen lazım. Her kavramın altındaki `tr:` ipucu anlamı Türkçe pekiştirir. Sırayla oku; formülleri yüksek sesle tekrar et.

How it is organized. Part I = energy fundamentals and formulas. Part II = carbon accounting, EPC, CRREM and EU regulation (the parts you said you don't know well — these are the most detailed). Part III = IoT, HVAC and fault detection. Part IV = the platform itself (architecture, the 9 report pages, the web app, the AI Copilot). Part V = the honesty layer (what is solid vs. what is indicative). Part VI = market and business. Part VII = EXIST, founder story, and a question bank.

The golden rule. Don't just memorize a number — know *why* it is that number. If asked "why a 5% discount rate?", the answer is "because it is a conservative real discount rate typical for commercial-real-estate investment appraisal", not "because it's in the code".

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2 PART I — ENERGY FUNDAMENTALS & FORMULAS

These are the metrics the report computes. Master the formula, the unit, and the reason it exists.

2.1 EUI — Energy Use Intensity (the master efficiency metric)

What it is. Energy consumed per unit of floor area per year. It is the single most important way to compare buildings of different sizes. In the ISO 50001 energy-management world the same idea is called an **EnPI (Energy Performance Indicator)**.

```
EUI = Annual energy consumption (kWh) / Conditioned floor area (m²)
Unit: kWh/m²/year
```

Why it exists. You cannot compare two buildings by raw kWh — a big building always uses more. Dividing by area turns consumption into *efficiency*, which is comparable. EPC certificates, EU targets and all sector benchmarks are expressed in kWh/m²/year.

> **tr:** EUI = yıllık kWh / m². Binaları "kaç kWh" diye değil, "m² başına ne kadar verimli" diye karşılaştırırsın. Tüm benchmark'lar bu birimde.

Conditioned vs. gross area. "Conditioned area" means the heated/cooled floor area — the space that actually consumes HVAC energy. Using gross area (including unheated garages, etc.) would understate intensity. EnergyLens stores both `gross_floor_area_m2` and `conditioned_area_m2`.

2.1.1 Climate adjustment (the part an energy academic will probe)

A building in Berlin burns more heating energy than an identical building in Istanbul — not because it is less efficient, but because it is colder. To compare fairly across climates and across years, you **normalize by degree-days**.

```
Climate-adjusted EUI = EUI × (Reference HDD+CDD / Actual HDD+CDD)
```

- **HDD — Heating Degree Days:** how much, and for how long, the outside temperature was below a base at which heating is needed. `HDD = Σ over days of max(0, T_base_heat − T_daily_avg)`, with **T_base_heat = 15 °C**.
- **CDD — Cooling Degree Days:** the mirror image for cooling. `CDD = Σ over days of max(0, T_daily_avg − T_base_cool)`, with **T_base_cool = 22 °C**.
- The base temperatures (15 °C / 22 °C) follow **EN ISO 15927-6**, the European degree-day standard.

Worked example. A building uses 150 kWh/m²/yr in a year with 3,000 HDD+CDD. The reference climate is 2,500 HDD+CDD. Climate-adjusted EUI = 150 × (2,500 / 3,000) = **125 kWh/m²/yr** — i.e. once you remove the "it was a harsh year" effect, the building actually performs better than the raw number suggests.

> **tr:** İklim düzeltmesi = tüketimi derece-güne göre normalize etmek. Soğuk yıl/şehir cezalandırılmasın diye. **Mutlaka bil:** EnergyLens hem HDD hem CDD kullanır (ratio metot, referans baseline ile). Eskiden sadece HDD'ye bölüyordu — o bir hataydı, düzeltildi.

2.1.2 Office benchmarks (know the bands)

Class	EUI (kWh/m²/yr)	Meaning
Excellent	< 80	LEED-Platinum-like
Good	80–130	Above sector average
Average	130–200	Typical commercial

Class	EUI (kWh/m²/yr)	Meaning
Poor	> 200	Urgent action

Different building types have different bands: a **hospital or data center is naturally far higher** (continuous loads, ventilation), a warehouse far lower. EnergyLens applies building-type-aware thresholds (e.g. Hotel/Healthcare ~130, Logistics ~60 as the monitoring trigger).

> **tr:** Benchmark bina tipine göre değişir — hastane/veri merkezi doğal olarak yüksek, depo düşük. “200 üstü kötü” sadece ofis için.

2.2 Peak demand, load factor and base load (the demand-side trio)

2.2.1 Peak demand

What it is. The highest instantaneous power draw (kW) in the period.

```
Peak demand (kW) = MAX(demand_kw) over the period
```

Why it matters — demand charges. Commercial electricity bills have two parts: an *energy* charge (€/kWh for total consumption) and a **demand charge** (€/kW/month for the single highest 15-minute average power). One bad spike at 10:00 on one morning can set the demand charge for the **entire month**. This is often the largest controllable line on a commercial bill — and exactly what a battery’s “peak shaving” attacks.

```
If a battery is present:
Peak shaved (kW)        = Peak_without_battery − Peak_with_battery
Demand-charge saving €  = Peak_shaved_kW × demand_charge_rate (€/kW/month)
```

> **tr:** Demand charge = aylık en yüksek 15-dk kW üzerinden alınan kapasite ücreti. Ticari faturanın en büyük kalemlerinden. Batarya tam bunu tıraşlar (peak shaving).

2.2.2 Load factor

What it is. How “flat” the consumption is — average power divided by peak power.

```
Load factor = Average demand (kW) / Peak demand (kW)    (range 0–1)
```

1.0 = perfectly flat (ideal); 0.3 = very spiky (you pay high demand charges and your equipment short-cycles). Office target **> 0.70**. A low load factor is a money-loss signal even if total kWh looks fine.

> **tr:** Load factor tüketimin ne kadar “düz” olduğunu ölçer. Düşükse = tarife mahkumu + ekipman yıpranması. Hedef >0.70.

2.2.3 Base load (sleep consumption)

What it is. The minimum power the building draws when it should be empty (02:00–04:00).

```
Base load (kW) = MIN(demand_kw) during 02:00–04:00 local time
Alert if: Base load / Peak demand > 0.35
```

If a building still pulls more than ~35% of its peak in the middle of the night, something is not shutting down — servers, lighting, HVAC scheduling, or stubborn standby loads. This is one of the cheapest savings to find.

> **tr:** Gece 2–4 bina "uyumalı". Hâlâ tepenin %35'inden fazla çekiyorsa bir şey kapanmıyor. En ucuz tasarruf kalemlerinden.

2.3 Solar PV (active when a building has photovoltaics)

Solar is measured by four numbers. The two that clients always confuse are the first two — know the difference cold.

```
Self-Consumption Rate  = self_consumed_kwh / generated_kwh       target > 70%
Self-Sufficiency Rate  = self_consumed_kwh / total_consumption_kwh  typical 20–
60%
Solar Yield            = generated_kwh / pv_capacity_kwp         (kWh/kWp/yr)
Performance Ratio (PR) = actual_yield / theoretical_yield       target > 0.75
   where theoretical_yield = irradiance_kwh_m² × pv_capacity_kwp × 0.80
```

- **Self-Consumption Rate** answers *"of the solar I generated, how much did I use myself?"* (vs. exporting to the grid). Low value → you are exporting cheaply; a battery or load-shifting would help.
- **Self-Sufficiency Rate** answers *"of my total need, how much came from my own solar?"* (vs. importing from the grid). These are **different fractions with different denominators** — mixing them up is the classic beginner error.
- **Solar Yield**: Germany ~900–1,050 kWh/kWp/yr; Turkey ~1,300–1,600 (sunnier). A quick sanity check on whether an array is sized/sited well.
- **Performance Ratio (PR)** is the health metric: actual output vs. what the sunlight *should* have produced. Industry default assumption 0.80. PR < 0.75 → panel soiling, shading, or inverter fault.

> **tr:** İki kavramı ayır: **Self-Consumption** = "ürettiğimin ne kadarını kullandım" · **Self-Sufficiency** = "ihtiyacımın ne kadarını solar'dan karşıladım". Farklı paydalar! PR = panelin sağlık metriği.

kWp vs kWh. kWp (kilowatt-peak) is the array's *rated capacity* (a size). kWh is *energy produced* (a flow). 1 kWp of modern panels needs roughly **7 m²** of roof (400 W panels, ~1.7 m² each) — EnergyLens uses `max_kwp ≈ roof_area_m² / 7` to auto-suggest a system size.

2.4 Battery storage (active when a building has a battery)

2.4.1 Battery health metrics

```
Round-Trip Efficiency = discharged_kwh / charged_kwh     target > 0.90 (LFP)
Cycle count (daily)     = charged_kwh / capacity_kwh            (full-
equivalent cycles)
Usable SoC band       = 20%–80% recommended (15%–90% hard limit)
```

- **Round-trip efficiency**: of the energy you put in, how much you get back. LFP ~0.90–0.92. The rest is lost as heat.
- **State of Charge (SoC)**: charge level 0–100%. **Why the 20–80% band?** Below ~15% the cell chemistry degrades; above ~90% the electrodes are stressed. Staying in 20–

80% extends lifetime — the EU battery warranty (cycle durability) assumes disciplined cycling.

- **Cycles & lifetime**: LFP is rated ~4,000–6,000 cycles. Lifetime estimate = total cycles / guaranteed cycles.

  tr: Round-trip = "koyduğumun ne kadarını geri aldım" (>0.90). SoC bandı 20–80% çünkü altı/üstü hücreyi yıpratır → ömür uzar.

2.4.2 Battery chemistries (know the four)

Chemistry	Full name	Strength	Weakness	Where used
LFP	Lithium Iron Phosphate ($LiFePO_4$)	Safe, long cycle life, no cobalt	Lower energy density	Commercial stationary (recommended)
NMC	Nickel Manganese Cobalt	High energy density	Shorter cycle life, cobalt	EVs; common in Turkey
NCA	Nickel Cobalt Aluminium	Very high density	Cost, thermal sensitivity	Premium EV
Solid-State	Solid electrolyte	Future: density + safety	Not yet mass-market	Emerging

EnergyLens recommends **LFP** for commercial buildings (safety per **IEC 62619**, long life, cobalt-free).

> **tr:** Ticari bina için LFP öner — güvenli, uzun ömür, kobalt yok. NMC daha yoğun ama ömrü kısa (Türkiye'de yaygın).

2.4.3 The 7 dispatch strategies (when to charge/discharge)

"Dispatch" = the logic that decides, every interval, whether to charge or discharge. EnergyLens compares seven:

1. **Self-Consumption** — charge on solar surplus, discharge when load > solar. Goal: maximize self-consumption.
2. **Peak Shaving** — discharge when demand crosses a threshold (e.g. 90th-percentile of last 12 months) to cut the demand charge.
3. **Arbitrage / Time-of-Use** — charge when electricity is cheap, discharge when expensive (needs dynamic pricing).
4. **Time-of-Use optimization** — schedule around a fixed peak/off-peak tariff.
5. **Frequency Regulation** — sell fast response to the grid operator (advanced, needs market access).
6. **Demand Response** — reduce/shift load on a grid signal (links to the EU Demand-Response Network Code, ~2027).
7. **Hybrid** — blend of the above.

```
Self-consumption logic (every 15 min):
  net_load = consumption(t) - solar(t)
  if net_load < 0 (surplus):  if SoC < 90% charge else export
  if net_load > 0 (deficit):  if SoC > 15% discharge else import
```

```
Peak-shaving logic:
  if demand(t) > threshold:   discharge (demand − threshold), if SoC > 15%
  if demand(t) < threshold/2: charge (prefer off-peak tariff), if SoC < 80%
```

> **tr:** Dispatch = bataryayı ne zaman şarj/deşarj edeceğine karar veren mantık. En çok kullanılan ikisi: self-consumption (solar'ı kendine sakla) ve peak-shaving (demand charge'ı kır).

2.5 Heat pumps — COP / SCOP (active when a building has a heat pump)

What a heat pump is. A device that *moves* heat rather than *making* it, so it can deliver more heat energy than the electricity it consumes.

```
COP  = Heat energy produced (kWh) / Electricity consumed (kWh)
SCOP = Seasonal COP = season's total heat / season's total electricity
COP performance ratio = COP_actual / COP_rated
   < 0.80 → MEDIUM maintenance alert
   < 0.60 → HIGH critical-fault alert
```

A COP of 4 means 1 kWh of electricity delivers 4 kWh of heat — which is why heat pumps are central to decarbonizing heating. **SCOP** is the honest, season-averaged figure (COP varies hour to hour).

The nuance an energy academic will test: COP **falls naturally in cold weather** (bigger temperature lift). So before alarming on a low COP, you must ask "did the weather change?" EnergyLens compares COP **within equivalent outdoor-temperature bands**, so a normal winter dip is not mistaken for a fault.

> **tr:** COP = üretilen ısı / harcanan elektrik (4 = 1 kWh elektrikle 4 kWh ısı). **Kritik:** COP soğukta doğal düşer → alarmı eşdeğer dış-sıcaklık bandında yorumla, kışı arıza sanma.

Heat pump vs. gas boiler (the simulation logic).

```
Gas boiler cost  = heating_kwh / 0.90 (boiler efficiency) × gas_price
Heat pump cost   = (heating_kwh / COP_rated) × electricity_price
Saving           = gas_cost − hp_cost
```

In Germany, installing a heat pump **automatically satisfies the GEG §71 "65% renewable heating" rule** and unlocks the KfW heat-pump subsidy (see Part II).

2.6 Building envelope — U-value, WWR, air-tightness

The "envelope" is the building's skin (walls, roof, floor, windows). How much heat leaks through it drives the heating/cooling demand.

- **U-value (W/m²K)** — heat-transfer coefficient of a building element. **Lower = better insulation.** German GEG minimums: wall ≤ 0.24, roof ≤ 0.20, window ≤ 1.30 W/m²K. (A single-glazed window can be ~5; triple glazing ~0.8.)
- **Window-to-Wall Ratio (WWR)** — window area ÷ exterior wall area. More glass = more daylight but more heat loss/gain.

- **Air-tightness (ACH — Air Changes per Hour)** — how many times per hour the indoor air leaks out and is replaced uncontrolled. High ACH = wasted heat. Measured with a **Blower Door Test (EN 13829 / ISO 9972)**.
- **Thermal mass** — the building's ability to store heat (concrete = high, lightweight = low), which buffers temperature swings.

```
Insulation-upgrade saving:
  Q_saved (kWh) = ΔU × surface_area × HDD × 24 / 1000
  where ΔU = current_U − target_U  (W/m²K)
```

> **tr:** U-value = ısı kaçağı katsayısı (W/m²K), **düşük=iyi**. WWR=cam/duvar oranı. ACH=saatlik hava sızıntısı. Blower Door Test ile ölçülür. Bunlar ısıtma/soğutma yükünü belirler.

The A7 anomaly (insulation degradation). If a building's climate-adjusted EUI rises > 15% year-over-year while the weather and HVAC are unchanged, the envelope may be degrading (insulation settling, increased air infiltration) — EnergyLens flags it and recommends a Blower Door Test.

2.7 Carbon and cost

2.7.1 Carbon

```
CO2 from consumption (kg) = consumption_kwh × grid_emission_factor (kg CO2/kWh)
CO2 avoided (kg)       = solar_generated_kwh × grid_emission_factor  (grid power NOT imp
CO2 net (kg)               = CO2_consumption − CO2_avoided
Carbon intensity           = CO2_net / conditioned_area   (kg CO2/m²/yr)
```

Grid emission factor = how much CO_2 a kWh of grid electricity carries; it depends on the country's generation mix and falls as the grid gets greener. EnergyLens uses **year-indexed, sourced** factors from one reference table:

- **Germany (UBA / Umweltbundesamt):** ~0.43 (2022) → ~0.386 (2023) → ~0.363 (2024) kg CO_2/kWh, and still falling (~0.33 in 2025). The official annual factor for 2023 is ≈ 380 g/kWh.
- **Turkey (TEİAŞ):** 0.442 kg CO_2/kWh (single canonical value).
- Other countries: IEA placeholders, clearly labelled.

> **tr:** Grid emission factor = 1 kWh şebeke elektriğinin taşıdığı CO_2. Ülkeye + yıla göre değişir, şebeke yeşilleştikçe düşer. DE ~0.36–0.39 (UBA, yıllık), TR 0.442 (TEİAŞ). EnergyLens tek referans tablosundan, kaynak+yıl ile okur.

EU ETS & carbon credits. The EU Emissions Trading System is the EU carbon market. The price (mid-2026) is roughly **€75–77 per tonne CO_2** (2025–26 range ~€60–95). Avoided emissions have a monetary value:

```
carbon_credit_value € = CO2_avoided_kg / 1000 × EU_ETS_price (€/tonne)
```

2.7.2 Cost

```
Energy cost € = grid_import_kwh × energy_price
              + peak_demand_kw × demand_charge (€/kW/month)
              − grid_export_kwh × feed_in_tariff
```

Canonical assumptions: DE electricity ≈ 0.226 €/kWh (Eurostat non-household), gas ≈ 0.07 €/kWh, gas CO_2 factor 0.201 kg/kWh (DEFRA, single basis), PV feed-in ≈ 0.08 €/kWh (EEG, small commercial).

> **tr:** Maliyet = enerji ücreti + demand charge − şebekeye satıştan gelen feed-in. Üç parça. Çoğu kişi sadece kWh×fiyat sanır; demand charge'ı unutma.

2.7.3 ESG composite score (the 0–100 number on Page 6)

A platform-defined blend, **not** an official standard — present it as "our internal benchmark score":

```
ESG score = 40% (EUI vs sector benchmark)
          + 30% (CO₂ intensity vs country average)
          + 20% (renewable share / solar self-sufficiency)
          + 10% (compliance status)
```

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3 PART II — CARBON ACCOUNTING, EPC, CRREM & EU REGULATION

This is the part you said you don't know well. It is also the part a sustainability-minded academic cares most about, and the part that gives the platform its commercial value. Take it slowly.

3.1 GHG Protocol — Scope 1, 2, 3 (the foundation)

The **GHG Protocol** (by the World Resources Institute + WBCSD) is the global **methodology** for counting greenhouse-gas emissions. EU rules (ESRS E1) *reference* it. It splits emissions into three "scopes":

- **Scope 1 — Direct emissions** from sources you own or control. For buildings: **gas boilers, company vehicles, and refrigerant leaks** from HVAC (refrigerants like R-410A have very high global-warming potential and are routinely forgotten).
- **Scope 2 — Indirect emissions from purchased energy** — the grid electricity, district heat, steam or cooling you buy. **Reported two ways, and both are required under ESRS E1-6:**
  - *Location-based*: uses the country grid-average emission factor.
  - *Market-based*: uses your specific supplier's factor (e.g. a green-electricity contract). Companies routinely forget market-based and get flagged in audit.
- **Scope 3 — All other value-chain emissions** — 15 categories. For buildings the material ones are usually **Category 13 (downstream leased assets — i.e. tenant energy use)** and **Category 1 (purchased goods / embodied carbon of materials)**.

> **tr:** Scope 1 = senin yaktığın (gaz, araç, soğutucu gaz kaçağı). Scope 2 = satın aldığın elektrik (hem location- hem market-based raporla!). Scope 3 = değer zincirindeki her şey (15 kategori; binalarda en büyüğü genelde kiracı tüketimi = Kat. 13).

How EnergyLens computes them (and the honest limits): - Scope 2 = `net_grid_kwh × year-indexed emission factor` (location-based, live). Market-based mechanism exists via a supplier-factor column but needs real contract data to differ. - Scope 1 = from a gas meter if present; otherwise proxied from a heating share of electricity (a *screening* estimate). Refrigerant leakage is **not yet** included. - Scope 3 = `(Scope 1 + Scope 2) × 8%`, explicitly flagged `scope3_disclosure_grade = False`. This is a **screening placeholder, not a disclosure-grade number** — say so plainly.

> **tr: Dürüst ol:** Scope 3 şu an %8 screening (disclosure-grade DEĞİL diye etiketli); market-based S2 mekanizması var ama gerçek tedarikçi verisi bekliyor; refrigerant Scope 1 henüz yok. Bunları "indicative, roadmap'te güçlendiriyorum" diye söyle.

3.2 EPC — Energy Performance Certificate

What it is. A national rating, **A+ to G**, of a building's energy performance (based on kWh/m²/yr), required across the EU at sale or rent. Each country has its own methodology (Germany: issued under the **Energieausweis** system).

The area-weighting point (an "audit failure" to avoid). A portfolio's EPC score must be **weighted by floor area**, not a plain average of buildings — otherwise a tiny building counts the same as a huge one. The correct calculation:

```
Portfolio EPC = Σ(area × score) / Σ(area)     [area-weighted, correct]
          ≠  average(score)                 [plain average, an audit failure]
```

EnergyLens computes the area-weighted version (it previously used a plain average — that was caught in self-audit and fixed).

> **tr:** EPC = A+→G ulusal enerji notu. Portföy skoru **alan-ağırlıklı** olmalı (düz ortalama bir denetim hatasıdır — 50 m² ile 50.000 m² eşit sayılamaz).

3.3 CRREM — Carbon Risk Real Estate Monitor

What it is. A science-based tool that draws a **decarbonization pathway** — a curve of allowed carbon intensity (kg CO_2/m²/yr) that falls year by year to stay within 1.5 °C / 2 °C. A building is plotted against the curve for its type and region.

Stranding. The year a building's carbon intensity **crosses above** the pathway is its **"stranding year"** — the point at which it risks becoming a regulatory/market liability (an "stranded asset"). Investors and REITs care intensely about this.

```
carbon_intensity = (CO₂_last_30d × 365.25/30) / area
stranding year   = the year the pathway curve drops below carbon_intensity
```

Two honest notes: 1. EnergyLens uses **indicative** 1.5 °C pathway anchors, not the official CRREM v2 dataset — embedding the official curves requires a **CRREM License Partner** agreement. The curves are isolated in one module so they can be swapped for licensed values later. 2. The comparison must be on the **same scope**: the building's Scope 1+2 intensity vs. a Scope 1+2 pathway (CRREM excludes Scope 3). EnergyLens was corrected to compare on this basis.

> **tr:** CRREM = bilimsel dekarbonizasyon eğrisi; binanın yoğunluğu eğriyi hangi yıl aşarsa o yıl "stranded" (atıl varlık riski). REIT'ler buna bayılır. Bizimkiler **indikatif** (resmî değil — License Partner sonra). Karşılaştırma Scope 1+2 bazında.

CRREM timing tip — tr: CRREM'in ücretsiz Excel aracı **1 Temmuz 2026'da emekliye ayrılıyor** → bizim entegre, sürekli güncellenen yaklaşımımız tam zamanında değer kazanıyor.

3.4 EU & national regulations (current as of mid-2026)

Learn each as: **what it is · who it binds · the key dates/numbers · how EnergyLens supports it.** These dates are current — verify again right before the meeting if you want to be safe.

3.4.1 CSRD — Corporate Sustainability Reporting Directive

- **What:** the EU law requiring companies to publish sustainability reports using the **ESRS** standards. It replaced the older NFRD.
- **The 2026 "Omnibus" change (important — the scope shrank):** the Omnibus simplification directive was published in the EU Official Journal on **26 February 2026** and entered into force on **18 March 2026**. It **narrows scope** to EU companies with **> 1,000 employees AND > €450 million net turnover**. Listed SMEs are now fully exempt; ESRS data points are being cut substantially.
- **How to position EnergyLens:** because mandatory scope shrank, **do not say "everyone must comply."** Say **"report-ready / ESRS-E1-aligned reporting support"** — smaller firms can still report voluntarily (the VSME standard), and larger ones still need the data.

tr: CSRD = AB sürdürülebilirlik raporlama yasası (ESRS kullanır). **2026 Omnibus** kapsamı daralttı: >1000 çalışan & >€450M ciro. Sen "herkese zorunlu" DEME — "report-ready / ESRS-E1-aligned" de.

3.4.2 ESRS E1 — the Climate Change standard

- **What:** the most important of the 12 ESRS topical standards. It has **9 disclosure requirements, E1-1 to E1-9.** The four to memorize:

Code	Discloses	EnergyLens page
E1-1	Climate transition plan	Pages 4–5 (forecast, decision support)
E1-5	Energy consumption & mix	Pages 1–3
E1-6	Gross Scope 1, 2, 3 + total GHG	Page 6 + /compliance ESRS export
E1-9	Anticipated financial effects	Page 9 (battery ROI, scenarios)

tr: ESRS E1 = iklim standardı, 9 açıklama (E1-1...E1-9). Şu eşleştirmeyi ezberle: E1-6 = Scope 1/2/3 = bizim Sayfa 6. Hoca "E1-6 yapabiliyor musun?" derse kafanda Sayfa 6 belirsin.

3.4.3 EPBD — Energy Performance of Buildings Directive (recast 2024)

- **What:** the EU directive setting building energy standards. The **recast Directive (EU) 2024/1275** entered into force **28 May 2024**, with national transposition due **29 May 2026**.

- **MEPS (Minimum Energy Performance Standards) for non-residential buildings — the dates an academic will check:** member states must set national MEPS by **1 January 2027**; the worst-performing **16%** must be renovated by **2030** and **26%** by **2033** (worst non-residential buildings reaching at least class **F by 2030** and class **E by 2033**). New buildings must be **zero-emission by 2030**.
- **How EnergyLens supports it:** the `/compliance` MEPS Radar bands buildings by EPC against these milestones (indicative, since national thresholds aren't all final yet).

tr: EPBD = AB bina direktifi (2024 recast). Ticari MEPS: ulusal kurallar 2027, en kötü %16 → 2030 (en az F sınıfı), %26 → 2033 (E). Yeni binalar 2030 zero-emission. Bizim MEPS Radar bunu tarar (indikatif).

3.4.4 EnEfG — German Energy Efficiency Act (Energieeffizienzgesetz)

- **What:** requires organizations above an energy-use threshold to run **energy-management systems**. Companies with **≥ 250 employees** (or high energy use) need **ISO 50001 certification or an equivalent energy audit**; companies with annual final energy consumption above **7.5 GWh** must implement energy- or environmental-management systems and act on economical efficiency measures.
- **EnergyLens:** checks `employee_count` and `iso50001_certified`, and flags the gap with a recommended action (and matches the BAFA energy-audit subsidy).

tr: EnEfG = Alman enerji verimliliği yasası. ≥250 çalışan → ISO 50001 veya denetim zorunlu; >7.5 GWh tüketim → enerji yönetim sistemi. Biz bunu kontrol edip eksikse uyarı + teşvik eşleştirme veriyoruz.

3.4.5 GEG — German Building Energy Act (Gebäudeenergiegesetz)

- **What:** the law governing building energy use and heating. **§71:** heating systems installed from 2024 must use **≥ 65% renewable energy** — which a **heat pump satisfies automatically**. It also sets minimum **U-values** (wall ≤ 0.24, roof ≤ 0.20, window ≤ 1.30 W/m²K).
- **EnergyLens:** ties the heat-pump recommendation to GEG §71 and flags non-compliant envelopes.

tr: GEG = Alman bina enerji yasası. §71: 2024'ten sonraki ısıtmada ≥%65 yenilenebilir (ısı pompası otomatik karşılar). Minimum U-value'lar. Biz ısı pompası önerisini buna bağlıyoruz.

3.4.6 EEG, EU Taxonomy, EU Battery Regulation, BEP-TR

- **EEG (Renewable Energy Sources Act, Germany):** sets the **feed-in tariff** the grid pays for exported PV (~€0.08/kWh, small commercial).
- **EU Taxonomy:** a classification of which economic activities count as "environmentally sustainable" for investor reporting. Relevant building activity is **7.7 "Acquisition and ownership of buildings."** EnergyLens gives an **indicative "alignment indication"** for the climate-mitigation substantial-contribution criterion only — never claim full "Taxonomy-aligned" (that also needs DNSH + minimum safeguards).
- **EU Battery Regulation — the correct number is Regulation (EU) 2023/1542** (NOT "2023/1670" — see the correction box below). It requires, for batteries sold

in the EU: a **carbon-footprint declaration**, recycled-content disclosure, cycle-durability/warranty info, and a **digital Battery Passport via QR code from 18 February 2027** (for industrial/EV/LMT batteries > 2 kWh); the industrial carbon-footprint declaration applies from **18 February 2026**. EnergyLens only puts EU-compliant chemistries into Page 9 scenarios.
- **BEP-TR (Turkey — Binalarda Enerji Performansı):** requires an EPC, renewed every 10 years.

CORRECTION — tr (ÖNEMLİ): App ve dokümanlarda “EU 2023/1670” yazıyor ama AB Batarya Regülasyonu’nun doğru numarası **2023/1542**’dir. Hocaya **2023/1542** de. Bunu app’te de düzeltmen iyi olur (yakalanmamak için). Battery Passport: 18 Şubat 2027 (QR), endüstriyel karbon beyanı 18 Şubat 2026.

3.5 Incentives / subsidy programs (what the recommendation engine matches)

When EnergyLens recommends an action, it matches a funding program. Know these:

- **KfW (Kreditanstalt für Wiederaufbau)** — German state development bank. Runs **BEG (Bundesförderung für effiziente Gebäude)** building-efficiency grants, e.g. **KfW 261** for deep renovation. **Since 1 Jan 2025, KfW also handles heat-pump funding.**
- **BAFA (Bundesamt für Wirtschaft und Ausfuhrkontrolle)** — runs building-envelope measures and the **energy-audit** subsidy.
- **Heat-pump subsidy structure (2025–26, residential — use as the illustrative model):** base **30%** + **5%** efficiency bonus (natural refrigerant / geothermal) + **20%** “climate-speed” bonus (replacing a fossil system > 20 years old) + **30%** income bonus (household income < €40k) → **up to 70%**, on a maximum eligible **€30,000/unit → up to ~€21,000 grant**. Applications via the “Meine KfW” portal.
- **YEKA (Turkey — Yenilenebilir Enerji Kaynak Alanları):** renewable-energy zones / PV incentive scheme.

tr: KfW = devlet kalkınma bankası (BEG/261 bina verimliliği; 2025’ten ısı pompası da). BAFA = kabuk önlemleri + enerji denetimi. Isı pompası teşviki: %30 baz + bonuslar → %70’e, ~€21k’ya kadar. Biz öneriyi uygun teşvikle eşleştiriyoruz.

3.6 Technical standards (recognize and name-drop accurately)

Standard	What it defines	Where in EnergyLens
ISO 50001	Energy Management Systems; defines EnPI & EnB	EnEfG compliance check
EN ISO 15927-6	Degree-day method (HDD/CDD base temps)	Climate-adjusted EUI
EN 16798-1	Indoor comfort parameters (20 °C heat / 24 °C cool setpoints)	HVAC setpoint logic
ASHRAE 90.1	Building energy-efficiency benchmarks	EUI bands
ASHRAE 135	BACnet protocol	IoT adapter (P0)

Standard	What it defines	Where in EnergyLens
ASHRAE Guideline 36	High-performance HVAC control & AFDD	Page 8 fault detection
IEC 61158	Modbus protocol	IoT adapter (P0)
IEC 62619	Industrial Li-ion battery safety	LFP recommendation
EN 13829 / ISO 9972	Blower-door air-tightness test	A7 anomaly action

> **tr:** Bu standartları "tanı ve doğru telaffuz et" düzeyinde bil. Hoca birini sorarsa ne işe yaradığını + EnergyLens'te nerede kullanıldığını söyle. EnPI/EnB = ISO 50001 terimleri (EnB = Energy Baseline, ölçüm referansı).

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4 PART III — IoT, HVAC ANALYTICS & FAULT DETECTION

This is the platform's strongest, most sector-grade layer (Page 8). Be ready to speak about it with confidence.

4.1 IoT protocols — why support several

Building Management Systems (BMS) do not speak one language; the market is fragmented. Support one protocol and you lose half the buildings. So EnergyLens uses a **protocol-adapter pattern**: each protocol has an adapter class that normalizes raw readings into standard units (°C, %, ppm, kW, kWh).

Protocol	Standard	Note	Priority
BACnet/IP	ASHRAE 135	The German BMS standard	P0
Modbus TCP	IEC 61158	Very common in EU industry	P0
MQTT 5.0	ISO/IEC 20922	Lightweight pub/sub, scalable	P1
REST API	HTTP/JSON	Catch-all	P1
OPC-UA	IEC 62541	Premium industrial automation	P2

> **tr:** BMS'ler tek dil konuşmaz → tek protokol = yarım pazar. Bu yüzden her protokol için bir adapter (hepsi standart birime çevirir). BACnet = Almanya standardı, Modbus = AB endüstri. Bu çok-protokol desteği nadir bir güç.

4.2 Sensor types and the dynamic dimension

`sensor_type` is a **dimension, not a hardcoded list** — each building exposes only the sensors it actually has, and the visuals adapt. EnergyLens models ~31 sensor types, grouped: indoor air quality (CO_2, VOC, PM2.5/PM10, lux, dB), occupancy, energy sub-metering (with power factor), HVAC equipment telemetry, water, and PV/battery/EV.

The minimum set every building exposes: HVAC temperature, humidity, CO_2, total building power (kW), and HVAC-only power. The extended set (full BMS buildings) adds supply/return temperatures (delta-T efficiency), sub-metering (lighting, plug loads), chiller COP, boiler efficiency, pump pressure, fan RPM.

> **tr:** sensor_type sabit liste değil, **dimension** — her bina sadece kendi sensörlerini gösterir, görseller uyum sağlar. ~31 tip. Bu, "her binaya boş N/A hücreler" yerine temiz, ilgili gösterge demek.

4.3 AFDD — Automated Fault Detection & Diagnostics (ASHRAE Guideline 36)

This is the heart of Page 8. **ASHRAE Guideline 36** is the industry standard for high-performance HVAC sequences and automated fault detection. EnergyLens implements **12 diagnostic rules** across zones, air-handling units (AHU), chillers, boilers, pumps and the whole building, with:

- **Frozen fault codes** (a stable, versioned vocabulary so visuals never break),
- **Diagnosis-level output**: each fault carries an occurrence/persistence count, a confidence score, a priority score, a probable cause, and an estimated energy/€ impact,
- **Transient suppression** (deadbands and a persistence requirement of ≥ 2 intervals) so a momentary blip is not reported as a fault — exactly as Guideline 36 prescribes.

```
Anomaly cost estimate (always labelled "Est."):
 cost_eur = duration_hours × power_waste_kw × grid_price (€/kWh, from the tariff referen
  HVAC temp violation: 2–5 kW extra per °C of deviation
  CO₂ spike > 1500 ppm: 1–3 kW extra ventilation
  Power spike > 120% of baseline: the actual excess kW
```

> **tr:** AFDD = otomatik arıza tespiti + teşhis. **ASHRAE Guideline 36** sektör standardı. 12 kural, frozen kodlar, teşhis-düzeyi çıktı (occurrence + confidence + priority + probable cause + €etki). Deadband + persistence (≥2) ile anlık dalgalanmayı arıza saymaz. Bu, ticari BMS satıcılarının kullandığı standart — vurgula.

Honest scope note: the demo runs on a **static IoT snapshot** (sensor master + generated bronze readings + processing + AFDD) so it shows live even when the trial EventStream is off. Economizer / simultaneous-heating-and-cooling rules were **deliberately skipped** (no outdoor-air-temp / split-valve telemetry yet — no faking), planned for Phase 2.5.

> **tr:** Demoda statik snapshot canlı görünür (EventStream kapalı olsa bile). Bazı kuralları (economizer vb.) **kasıtlı atladım** çünkü gerekli telemetri yok — uydurmuyorum. Bu dürüstlük iyi bir sinyal.

4.4 HVAC analytics (Page 7)

The HVAC page reports heat-pump COP (actual vs rated), the **supply/return air temperature delta** (a proxy for HVAC efficiency — a healthy system moves a meaningful delta-T), runtime hours, and HVAC's share of total consumption.

The honest limitation to state up front: the heating/cooling/ventilation **split is modeled**, not sub-metered. It is derived from a building-type coefficient times the weather shape (HDD/CDD), so an Office's HVAC share is essentially pinned by its type. A facilities engineer will recognize this immediately, so frame the visual as *"modeled split using the HDD/CDD method — sub-metering refines this"* and offer sub-metering as a Tier-3 upsell.

tr: Sayfa 7'de HVAC split **modellenmiş** (sub-metered değil) — bina tipi katsayısı + hava şekli. HVAC mühendisi anlar. "Modeled split, sub-metering refines this" diye çerçevele.

4.5 Anomaly detection (Page 3) — the 7 rules

One authoritative engine writes the anomaly log. The seven rule families:

```
A1 Consumption spike  – consumption > 1.5× rolling avg (same hour & day-
type) AND weather not the cause   [MEDIUM]
A2 High base load    – base_load > 0.35 × peak, 01:00–05:00
A3 COP degradation   – COP < 0.80 × rated AND outdoor temp near design (not weather)
A4 Solar underperf.  – PR < 0.70 AND irradiance > 400 W/m²
A5 Battery anomaly   – round-trip < 0.85 / SoC stuck ≥95% / SoC < 10% repeatedly
A6 Weekend/holiday   – weekend consumption > 0.60 × weekday avg (HVAC not in holiday mode
A7 Insulation degrade – climate-adjusted EUI up > 15% YoY, weather & HVAC unchanged → Blow
Door test    [MEDIUM]
```

Each anomaly carries a severity, a probable cause, a recommended action, and an estimated € loss, in three languages (EN/DE/TR).

tr: 7 anomali ailesi. **Dürüst not:** COP/solar anomalileri proxy (gerçek COP sadece ısı pompalı binada); SOLAR_PR gürültüsü olabilir (üretim hâlâ sentetik). Önce sebebi (mevsim? hava?) ele, sonra alarmı yorumla.

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5 PART IV — THE PLATFORM

5.1 Microsoft Fabric — the stack (show your DP-600 mindset)

- **Microsoft Fabric** — Microsoft's unified analytics platform (GA May 2024) combining data engineering, warehousing, real-time analytics, data science and Power BI on one SaaS foundation.
- **OneLake** — one logical data lake for the whole organization; every workload reads/writes the same layer, so you stop copying data between systems. This single-copy lineage is *the* reason Fabric fits auditable carbon reporting.
- **Lakehouse** — storage that combines data-lake flexibility with warehouse structure (Delta Lake format on OneLake).
- **Medallion architecture** — **Bronze** (raw) → **Silver** (cleaned, conformed) → **Gold** (business-ready). Each layer builds on the previous; this is where row-level lineage comes from.
- **DirectLake mode** — Power BI reads the Delta/Parquet files *directly* from OneLake: no import, no DirectQuery. It gives import-speed performance with real-time freshness — the headline Fabric feature.
- **DirectQuery vs Import (the contrast):** Import = fast but stale (needs refresh); DirectQuery = live but slow. DirectLake gives the good half of both.
- **SQL Analytics Endpoint** — a SQL (ODBC/pyodbc) entry point to the Lakehouse. The app's custom React pages read Fabric here (SQL, not DAX).
- **EventStream + Eventhouse (KQL)** — Fabric's real-time ingestion + time-series database (queried with Kusto Query Language). The live path for Page 8 IoT.

- **Row-Level Security (RLS)** — restricts which rows (which buildings) a user sees; EnergyLens applies RLS on `building_id`.
- **Capacity (F-SKU)** — Fabric's compute/pricing unit (F2, F4, F8…); F2 is the smallest viable for a production proof-of-concept.

  tr: Fabric = tek SaaS analitik platformu. OneLake = tek veri gölü (kopyalama yok → lineage). Medallion = Bronze→Silver→Gold. DirectLake = import yok/DirectQuery yok → hız + tazelik. SQL Endpoint = app'in custom sayfaları buradan okur. RLS = kim hangi binayı görür.

5.2 End-to-end data flow

```
SOURCES  meters, bills, BMS sensors (BACnet/Modbus/MQTT), weather (Open-Meteo/DWD),
       prices (ENTSO-E), grid CO₂ (ElectricityMaps), solar & battery telemetry
      │   ingestion: Data Factory (batch, Tier 1) / EventStream (real-time, Tier 2/3)
   ▼
LAKEHOUSE (OneLake, Delta)   BRONZE → SILVER → GOLD
   │  DirectLake
   ▼
SEMANTIC MODEL (Power BI)  9-page report + DAX library (v57–v60)
   │
   ├─ Embedded Power BI (DirectLake)      ── what the building pages show
   ├─ Custom React (SQL Analytics Endpoint) ── what /portfolio reads
   └─ AI Copilot (LLM tool use → dispatcher) ── what /copilot uses
```

These **three parallel paths into Fabric** were all validated end-to-end — a sign of engineering maturity, and a hedge against any single path failing.

> **tr:** Fabric'e 3 paralel yol: (1) PBI DirectLake embed, (2) custom React → SQL Endpoint, (3) AI Copilot → tool use. Üçü de uçtan uca çalışıyor → tek noktaya bağımlı değilim.

5.3 The reference layer (single source of truth)

The biggest fix from the self-audit: emission factors and tariffs used to be scattered constants (the same Turkish factor appeared as 0.430 / 0.442 / 0.450 in four files). Now there is **one sourced, year-indexed reference layer** — `ref_grid_emission_factors`, `ref_electricity_tariffs`, `ref_fuel_factors` (each with source, URL, year) — and every engine joins it. This is the literal realization of "row-level lineage from a raw factor to a disclosed kg CO_2".

> **tr:** Denetimde en büyük açık: faktör/tarife dağınık sabitlerdi (TR 4 dosyada farklı). Çözüm: tek referans katmanı (kaynak+URL+yıl). Artık her motor oradan okur = lineage hikâyesi.

5.4 Tier-aware ingestion & three-layer access

- **Tiers:** Tier 1 (Insight) = batch, hourly, meters/bills. Tier 2 (Monitor) = + IoT, 5–15 min. Tier 3 (Copilot) = + battery/heat-pump telemetry, < 5 min. You don't pay for streaming when an hourly meter is enough.

- **Three-layer access** (how enterprise BMS products like Siemens Desigo / Schneider EcoStruxure work):
  - **Layer 1 — Power BI RLS (data):** which buildings a user sees.
  - **Layer 2 — App navigation (modules, Next.js):** which pages are unlocked (no IoT → Page 8 hidden; no battery → Page 9 locked).
  - **Layer 3 — Subscription tier (commercial):** which features the plan includes. RLS is always on — data never crosses a customer boundary.

  tr: Tier'a duyarlı veri alımı (gereksiz streaming ödemezsin). 3 katman erişim: RLS (veri) + app-nav (modül) + subscription (tier). Veri asla müşteri sınırını geçmez.

5.5 Data model & the 10 representative buildings

There is no real customer yet, so the platform runs on **10 representative buildings (B001–B010)** with **~3.5 years** of data, ~**693,000 data points**.

- **Why 10?** Enough diversity to demonstrate portfolio logic (comparison, ranking, RLS, area-weighting) without being unmanageable.
- **Why 3.5 years?** To capture seasonality (the HDD/CDD cycle), year-over-year trend (the A7 anomaly needs a prior year), and enough history to train the Prophet forecast.
- **Why ~693k points?** Hourly granularity × buildings × sensors × 3.5 years — so the demo feels production-scale, not a toy.
- **Diversity:** Germany, Turkey, Austria, Netherlands; Office/Retail/Hotel/Logistics/Healthcare/Data-Center/Lab; different technology profiles. Showcase buildings include B007 Copenhagen Net-Plus, B008 Leipzig Plattenbau, B009 Frankfurt Data Center, B010 Stockholm Lab.

The spine — `silver_building_master` carries every dimension that gates logic: identity, geography + climate zone (Köppen), gross/conditioned area, type, subscription tier, technology flags (`has_pv`, `has_battery`, `has_heat_pump`, …), PV/battery/heat-pump details, full envelope (U-values, WWR, ACH, thermal mass, insulation year), EPC and regulatory profile. **Technology-profile gating** means a building with `has_pv = False` simply never shows solar visuals — no empty "N/A" cells.

> **tr:** 10 temsili bina (henüz müşteri yok), 3.5 yıl, ~693k nokta — mevsimsellik + trend + forecast için. Omurga = silver_building_master (tüm boyutlar). Teknoloji bayrakları görselleri açıp kapatır (boş N/A yok).
>
> **Honesty note — tr:** B003/B005'in kimliği bazı notebook'larda çelişiyor (şehir/tip). Düzeltme listesinde. **Demoda o iki binayı şehir-ismiyle anma**, B001/B002/B007–B010'a odaklan.

5.6 The 9 Power BI report pages

Each page: the question it answers, the user, the key visuals/formulas, and the honest known limit.

Page 1 — Portfolio Overview. *"How is my portfolio doing this month?"* (Energy Manager / C-suite). Totals (kWh, cost €, CO_2) + per-building EUI heat map + worst-3 buildings vs type benchmark. Source: `gold_kpi_daily`. *Limit:* a monthly-trend aggregation needs the hourly-table → date relationship wired in Power BI.

Page 2 — Building Deep-Dive. *"What is this building doing now?"* (Facility Manager). 24-hour hourly demand curve, today's peak vs baseline, month-to-date cost, climate-adjusted EUI. Source: `gold_kpi_hourly` filtered by `building_id`. *Limit:* PR noise (real weather, synthetic generation).

Page 3 — Anomaly Detection. *"Where is waste / a fault?"* Anomaly counts by severity & type, table with €-loss and resolution status. Seven rule families (above). Source: `gold_anomaly_log` (one authoritative engine).

Page 4 — Forecasting. *"What will next month cost?"* 7/30/90-day Prophet forecast with a confidence band → forecast cost & CO_2. Regressors: HDD/CDD + occupancy. *Limit:* accuracy (MAPE) is currently **in-sample** — say "in-sample fit"; roadmap is rolling-origin cross-validation.

Page 5 — Decision Support. *"What should I do, in what order, what does it save?"* Recommendations ranked by `priority_score`, each with €/kWh/CO_2 savings, effort, payback, and a matched incentive (KfW/BAFA/YEKA/EEG). The multi-country regulatory engine (DE/AT/NL/TR/EU) is a real differentiator. *Limit:* the old "CSRD score" was renamed to `eu_taxonomy_score` / `csrd_disclosure_readiness` — CSRD is a disclosure directive, not a kg/m² threshold.

Page 6 — Sustainability (ESG / GHG) — the flagship reporting page. *"What is my footprint, am I report-ready?"* Scope 1/2/3 donut, YoY Scope bar, **area-weighted** EPC heat map, ESG score, carbon-credit potential. Source: `gold_ghg_scope`. ESRS E1 mapping (E1-5, E1-6). *Limits (state plainly):* Scope 3 = 8% screening (not disclosure-grade), market-based Scope 2 needs supplier data, CRREM indicative, refrigerant Scope 1 excluded. Say **"ESRS-E1-aligned reporting support,"** never "audited CSRD disclosure."

Page 7 — HVAC & Hourly Profile. *"Is my HVAC efficient?"* Heat-pump COP (actual vs rated), supply/return delta-T, runtime, HVAC share. *Limit:* the HVAC split is **modeled** (type coefficient × weather), not sub-metered — frame as "modeled split, sub-metering refines this."

Page 8 — Real-Time IoT Monitoring (FLAGSHIP). *"What's happening now, which fault burns money?"* Four KPI cards (real-time kW; zone-comfort %; CO_2 level; active alerts + est. € today); 24h power trend (building vs HVAC); sensor-uptime matrix (rows = zones, cols = sensor_type, dynamic); zone-setpoint compliance; FDD diagnostics. Architecture Event Hub → EventStream → KQL Eventhouse + Lakehouse. ASHRAE Guideline 36 AFDD, 12 rules. *Limit:* demo runs on a static IoT snapshot (EventStream off after trial).

Page 9 — Battery Dispatch & ROI. *"Should I invest in a battery — which, when does it pay back?"* (Energy Manager / CFO / investor). Battery-technology matrix (chemistry × country → fitness scores), 7 strategies compared, country pricing (ENTSO-E), payback / NPV (5% discount, 10-yr, with inflation + salvage) / IRR (Newton-Raphson), EU Battery Regulation **2023/1542** compliance flag. 12 countries × 8 chemistries × 7 strategies. *Limit (CFO page — be honest):* savings are an **upper bound** (all discharge valued at the peak tariff; demand saving proportional to energy, not peak-kW) → payback optimistic; roadmap prices discharge by the hour it displaces.

> **tr:** 9 sayfayı "hangi soru / hangi kullanıcı / hangi formül / bilinen sınır" diye bil. Flagship'ler: Sayfa 6 (Sustainability), Sayfa 8 (IoT), Sayfa 9 (Battery ROI). Her birinde dürüst sınırı önce sen söyle.

5.7 The web application (every module)

Stack: **Next.js (frontend) + FastAPI (backend) + Azure PostgreSQL**, three-provider auth (Microsoft Entra ID + Google + Email/Password with bcrypt). The Power BI report is a **locked, version-controlled template**; the app adds navigation, access control, custom pages, the Copilot, PDF export and audit logging.

- **Landing / Login / Onboarding** — branded entry; a mandatory 5-step wizard creates the first building (with solar basics).
- **/portfolio** — the home dashboard; reads Fabric via the SQL Analytics Endpoint; portfolio + solar KPI rows, buildings table, "data not connected" banner, glossary tooltips, Export PDF.
- **/buildings, /buildings/[id], /buildings/compare** — list (persistent selection, sticky compare bar) → detail with the embedded Power BI report (service principal, V2 embed API, DirectLake, persistent in-place navigation) → compare. Module-locked deep links show a "locked" preview when a module isn't owned.
- **/solar** — dedicated solar page (custom SVG, no chart dependency).
- **/compliance** — the EU-agenda hub (positions us next to Schneider EcoStruxure / Siemens Desigo): **MEPS Radar · CRREM Stranding · Flexibility Readiness · EU Taxonomy · ESRS-E1 Summary** + Compliance PDF. All indicative, with an always-visible anti-greenwashing caveat.
- **/actions** — recommendations (Fabric read) + a Postgres status overlay (new/in-progress/done/dismissed); every change audited.
- **/alerts** — anomaly triage; `gold_anomaly_log` (read) × an acknowledge overlay; "unhandled" = unresolved & not-acked → nav badge.
- **/copilot** — the AI assistant (below).
- **/glossary** — 10 terms, single-source, surfaced as tooltips app-wide.
- **/settings** — org profile, members, invitations, Stripe subscription, activity (audit log).
- **/admin (founder-only)** — cross-org stats/tables + mutations (tier/status, module unlock, fabric link); all admin actions audited.
- **/billing, /pricing** — Stripe self-serve (4 tiers) + public pricing page.
- **/demo** — unregistered public access (6 sample buildings, RLS-restricted embed) — the marketing URL.
- **PDF export** — light-themed A4 report routes via `window.print()`, zero dependency.
- **Cross-cutting:** audit log on every mutation; graceful degradation (Fabric down → calm 503 notice, not a 500); branded 404/error/loading states; mobile-responsive; accessibility; SEO/OG.

  tr: App = PBI raporunu (kilitli şablon) sarmalayan katman: navigasyon + erişim kontrolü + custom sayfalar + Copilot + PDF + audit. /compliance hub'ı (5 AB özelliği) bizi Schneider/Siemens yanına koyuyor.

5.8 AI Copilot (the innovation argument)

A conversational assistant over the Fabric Lakehouse. The difference from most BI "copilots": it uses **tool use** — it runs *real queries* against the data — instead of summarizing a report page as text. And it can **act**, not just describe.

```
user question → orchestrator → LLM picks a tool → dispatcher runs it against Fabric/Postg
→ result returns to the LLM → LLM writes the answer → streamed to UI via Server-
Sent Events
(org/building RLS enforced on every tool call)
```

Tool	What it does
`query_kpi`	KPIs for a building (kWh, EUI, cost, CO_2, period delta)
`compare_buildings`	rank/compare buildings on a metric
`list_recommendations`	a building's recommendations by priority
`get_anomalies`	anomalies filtered by severity/type
`simulate_battery_scenario`	run a battery scenario (payback/NPV/IRR)
`update_action_status`	change an action's status (a **write**)

Provider abstraction: one environment variable switches Anthropic (Claude) / Azure OpenAI (GPT-4o) / Mock (currently active, because the Anthropic credit is €0). The Mock returns deterministic answers with *real* tool calls — production is one flag away.

> **tr:** Copilot tool use ile Lakehouse'a **gerçek sorgu** atar (metin özeti değil) ve aksiyon alabilir (update_action_status = yazma). 6 tool, her çağrıda RLS. Provider abstraction: Anthropic/Azure OpenAI/Mock — production'a geçiş tek env-variable. EXIST inovasyon argümanın bu.

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6 PART V — THE HONESTY LAYER (read twice)

With an energy academic, the biggest risk is defending an over-claim; the biggest win is naming the weak spot first, honestly, with a roadmap. That turns you from "salesperson" into "engineer." When a hard question comes: **(1) acknowledge it, (2) state what you do now and why (screening / indicative / modeled), (3) give the roadmap.** Never invent; "I don't know, but here's how I'd find out" beats a confident wrong answer.

6.1 What you can defend confidently (solid / fixed)

Medallion architecture + lineage; the **reference layer** (one sourced, year-indexed table for factors/tariffs); **climate-adjusted EUI** (ratio method, HDD+CDD); **area-weighted EPC**; **CRREM on Scope 1+2**; one authoritative anomaly engine; the multi-country regulatory engine; the battery financial-model *structure*; **Page 8 IoT/AFDD** (ASHRAE Guideline 36 — sector-grade); per-building Prophet forecast; real external data (Open-Meteo / ENTSO-E / ElectricityMaps); the AI Copilot (tool use, RLS, provider abstraction).

6.2 Known limits + honest framing (memorize the one-liners)

#	Limit	Say this
1	Scope 3 = 8% screening	"A screening estimate, flagged not disclosure-grade; roadmap models material categories from activity data."

#	Limit	Say this
2	Market-based S2 = location	"Mechanism is in place; equal until per-building supplier contract data is connected."
3	CRREM indicative	"Illustrative anchors; official curves need a CRREM License Partner agreement — isolated in one module to swap in."
4	HVAC split modeled	"Modeled HDD/CDD split, not sub-metered; sub-metering refines it (Tier-3)."
5	Battery ROI upper bound	"Model structure is correct; current savings value all discharge at the peak tariff — I'm refining to hour-by-hour, making payback more conservative."
6	Forecast MAPE in-sample	"In-sample fit now; rolling-origin cross-validation for a client figure."
7	Refrigerant Scope 1 excluded	"Routinely missed; a refrigerant log would close it."
8	Solar generation synthetic	"Weather is real; generation is synthetic for the demo — a real inverter feed cleans the PR signal."
9	EU Battery Reg. number	App says 2023/1670; the correct regulation is **2023/1542** — fix in the app.

> **tr:** Bu tabloyu ezberle. Her sınır için tek cümle dürüst çerçeven hazır olsun. Bu, bir akademisyenin gözünde en güçlü kozun.

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7 PART VI — MARKET & BUSINESS

7.1 The problem & why now

EU buildings (residential + non-residential) account for ~40% of EU energy consumption and ~36% of energy-related greenhouse-gas emissions — the single largest sector. Yet most portfolios still run on monthly bills and spreadsheets, blind to anomalies, regulatory risk and decarbonization options. Meanwhile EU regulation is tightening hard (EPBD/MEPS 2030/2033, EnEfG, GEG, CSRD/ESRS, the EU Battery Regulation). Owners face large, rising compliance exposure without modern tools.

> **tr accuracy note:** “40%/36%” rakamı **tüm binalar** için (sadece ticari değil). Hocaya “EU buildings account for ~40% of energy and ~36% of emissions” de — “commercial” deme, çünkü resmî istatistik tüm bina stokunu kapsar.

Why now (five tailwinds): binding EU climate regulation (2024–2033); the post-2022 energy-price shock made energy a board topic; Microsoft Fabric matured (GA 2024); IoT sensor costs collapsed; AI made pattern-recognition affordable.

7.2 Competitors

Competitor	Position	Gap EnergyLens fills
Measurabl	US enterprise (well-funded)	Mid-market EU
Aquicore	US offices	EU regulatory specifics
Cortexa	EU enterprise	Mid-market accessibility
BuildingIQ	Enterprise BMS	SaaS pricing, Fabric-native
Honeywell Forge	Enterprise, internal	Standalone product
Schneider EcoStruxure Resource Advisor	Peer (auditable reporting, benchmarking, AI Copilot)	We’re a peer; ours adds multi-protocol IoT + Fabric-native + mid-market price
Siemens Desigo CC / INSIGHT	BMS (building automation)	We’re the analytics/intelligence layer, not a BMS

One-line angle: *Microsoft Fabric-native + EU-regulation niche + mid-market pricing + an AI Copilot that reasons over the Lakehouse via tool use.*

> **tr:** Rakipler: Measurabl/Aquicore (US, enterprise), Schneider/Siemens (akran/BMS). Farkımız: Fabric-native + AB regülasyon nişi + mid-market + tool-use Copilot. Schneider’a karşı boşluk: çok-protokol IoT + denetlenebilir export + akran benchmarking.

7.3 Market size (TAM / SAM / SOM) & the software market

- **TAM (EU):** ~5 million commercial buildings × €100–700/building/month ≈ **~€18 B/year**.
- **SAM (DACH focus):** **~€4.3 B/year**.
- **SOM (3-year realistic):** 100–200 customers → **€1–2 M ARR**.
- **External market sizing (for credibility):** the **Building Energy Management Systems (BEMS) software market** is ~**US $8.4 B (2026)** growing to ~**$24 B by 2035** at ~**8.7% CAGR**; the broader energy-management software market is larger still. A growing market, not a niche in decline.

> **tr:** TAM ~€18B (AB), SAM ~€4.3B (DACH), SOM €1-2M ARR. Dış doğrulama: BEMS yazılım pazarı ~$8.4B (2026) → $24B (2035), %8.7 CAGR. Büyüyen pazar.

7.4 Pricing & unit economics

Tier	Price	Target
Insight	€99/building/month	SME offices
Monitor	€299/building/month	Mid-market property mgmt
Copilot	€699–1,500/building/month	Enterprise, healthcare, IoT-equipped
Portfolio Custom	€5k–50k/month	10+ building portfolios

CAC €500–1,500; LTV (3 yr) ≈ €54k; **LTV/CAC 36–108×**; payback 2–4 months; ~85% gross margin. First 3 months free (pilot) to win reference cases.

7.5 Go-to-market

Bottom-up, validation-first: start with the cheapest accessible customers (universities, mid-market property managers in DACH) where deals close in 2–8 weeks; use case studies to move up to hotels, healthcare and REITs; defer slow enterprise sales until the product is mature. Channels: direct LinkedIn outreach, the BSBI/Yaşar academic network, content marketing, and (later) the Microsoft Partner Network.

> **tr:** GTM = aşağıdan-yukarı, önce-doğrula. Ucuz/erişilebilir müşteriden başla (üniversite, DACH mid-market — 2-8 hafta kapanır), case study ile yukarı çık. Solo founder enterprise döngüsünü kaldıramaz; mid-market SaaS ekonomisi (€17k+ ARR/müşteri, %85 marj) işi çevirir.

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8 PART VII — EXIST, FOUNDER & QUESTION BANK

8.1 Founder (how to present yourself)

B.Sc. Energy Engineering (Yaşar) + M.Sc. Energy Management (BSBI, June 2026) + **Microsoft DP-600**. Built the entire stack — Fabric medallion, 9-page Power BI, IoT adapters, battery simulator, web app, AI Copilot — **solo, end-to-end, in ~5 weeks.** The rare combination is **energy domain depth × modern data engineering**, accelerated by AI-assisted development. 18-month Job-Search-Visa runway to incorporate (a German "UG").

8.2 EXIST Gründerstipendium — what you're asking for

- **What:** the German federal founder grant (~**€65k over 12 months** = €30k stipend + €30k material + €5k coaching), administered through a university.
- **What you need from Dr. Chelabi (mandatory):** an **academic mentor letter** (EXIST is not accepted without it) + a BSBI university commitment letter. You will draft the mentor letter for his review.
- **Timeline:** apply Jul–Aug 2026 → decision Sep 2026–Feb 2027 → start Q1 2027.
- **Acceptance boosters:** DP-600, the BSBI link + his positive reply, a ready pitch package, DACH-market depth, the EU-regulation niche, solo execution. A **BSBI campus pilot before August would raise odds ~20%.**

tr: EXIST = ~€65K / 12 ay (federal Almanya), üniversite üzerinden. Hocadan ZORUNLU: mentor letter (+ BSBI commitment). Taslağı sen hazırlarsın. BSBI pilotu kabul şansını ~%20 artırır.

8.3 Question bank (rehearse out loud — close the answer, then check)

1. **EUI & climate adjustment?** Annual kWh / conditioned m²; ratio method with HDD+CDD, reference baseline (EN ISO 15927-6).
2. **Scope 2 location vs market?** Both required (ESRS E1-6). Location live (year-indexed official factor); market-based mechanism present, needs supplier data.
3. **Scope 3?** 8% screening now, flagged not-disclosure-grade; roadmap = category-based activity data.
4. **CRREM stranding?** Carbon intensity vs type/region 1.5 °C pathway crossing = stranding year; pathways indicative (License Partner later); compared on Scope 1+2.
5. **COP alarm vs season?** Compare within equal outdoor-temp bands; <0.80 MEDIUM, <0.60 HIGH.
6. **Emission factors — consistent & sourced?** One reference table, source+URL+year; TR 0.442 (TEİAŞ), DE year-indexed (UBA ~0.36–0.39).
7. **Real or synthetic data?** Weather/price/grid-CO_2 real (Open-Meteo/ENTSO-E/ElectricityMaps); building consumption/solar synthetic (no customer yet).
8. **Forecast accuracy?** Prophet, per building, HDD/CDD+occupancy; MAPE in-sample now, cross-validation on roadmap.
9. **Is the IoT real?** Yes, flagship — ASHRAE Guideline 36 AFDD, 12 rules, ~31 sensor types, multi-protocol; static snapshot in demo, EventStream in production.
10. **Battery ROI realistic?** Structure correct (NPV/IRR/payback); current savings an upper bound; refining to hour-by-hour tariff matching.
11. **Why Microsoft Fabric?** OneLake single-copy lineage (audit-ready), DirectLake speed+freshness, DP-600 certified.
12. **Does the Copilot really read data?** Tool use — real queries via 6 tools, RLS on each; not text summaries; can also act (a write tool).
13. **Data isolation?** Three layers: RLS (data) + app-nav (modules) + subscription (tier); data never crosses a customer boundary.
14. **Competitors / differentiation?** Measurabl/Aquicore (US enterprise), Schneider/Siemens (peer/BMS); ours Fabric-native + EU niche + mid-market + tool-use Copilot.
15. **Market size?** TAM ~€18B (EU), SAM ~€4.3B (DACH); BEMS market ~$8.4B→$24B by 2035.
16. **How did you build this solo?** Energy M.Sc. × DP-600 × AI-assisted development; 5 weeks end-to-end.
17. **What exactly do you want from me?** The EXIST academic mentor letter (+ BSBI commitment), and optional monthly/quarterly advisory.
18. **What's in it for BSBI?** A free 6-month, RLS-secured campus pilot I run end-to-end; M.Sc. capstone + a named case study.
19. **What's next / 12 months?** BSBI pilot → 5 paying customers → incorporate (UG) → €5–10k MRR; strengthen Scope 3 / market-based S2 / CRREM official curves / sub-metering.
20. **EU Battery Regulation number?** 2023/1542 (correct the app's "2023/1670"); battery passport 18 Feb 2027.

tr: Bilmediğin soruda: "That's a great question — I don't have a confident answer right now, but here's how I'd find out..." Uydurmaktan her zaman iyidir.

End of the single-source reference. Study it twice, stay calm, be honest. You’re ready. — EnergyLens prep